

Compositional Simulation for Sequestration of CO₂ and Other Greenhouse Gases in Saline Aquifers

The modeling of CO₂ storage in saline aquifers involves the solution of the component transport equations, the equations for thermodynamic equilibrium between the gas and aqueous phase, and the equations for geochemistry. The latter involve reactions between the aqueous species and mineral precipitation and dissolution.

The modeling of transport of reactive components in the aqueous phase has been a subject of intensive research in the hydro-geochemical field ([Lichtner, 1988](#), [Lichtner, 1991](#), [Lichtner, 1992](#), [Liu and Ortoleva, 1996\(a\)](#), [Liu and Ortoleva, 1996\(b\)](#), [Sevougian et al, 1992](#), [Sevougian et al, 1993](#), [Steefel and Lasaga, 1992](#), [Steefel and Lasaga, 1994](#), [Steefel and MacQuarrie, 1996](#), [Yeh and Tripathi, 1989](#), [Yeh and Tripathi, 1991](#), and [Pruess et al, 2003](#)). There are essentially two approaches for solving the coupled system of equations: the sequential solution method and the simultaneous solution method. In the sequential solution approach, the flow equations and chemical-equilibrium equations are solved separately and sequentially. Iterations are applied between the two systems until convergence is achieved. The simultaneous solution approach solves all equations simultaneously with Newton's method. The simultaneous solution approach is also referred to as the fully-coupled approach. Although the fully-coupled method is recognized as the most stable approach, it has not been fully exploited as the required numerical solution method is very involved.

This Appendix describes the methods implemented in GEM for modeling of CO₂ storage in saline aquifers: adaptive-implicit multiphase multicomponent flow simulation with phase and chemical equilibrium and rate-dependent mineral dissolution/ precipitation using the fully-coupled approach. The success of this approach depends strongly on efficient sparse solution techniques and robust solution methods for multicomponent multiphase flow systems that were developed previously for an EOS compositional simulator ([Nghiem and Rozon, 1988 and 1989](#), and [Collins et al, 1992](#)).